
KTU
NOTES

MODULE - III

COMPLEX VARIABLE - DIFFERENTIATION

TOPICS:

Complex function, limit, continuity, derivative, analytic functions, Cauchy-Riemann equations, harmonic functions finding harmonic conjugate, conformal mappings - mappings $w = z^2$, $w = e^z$. Linear fractional transformation $w = \frac{1}{z}$, fixed points. Transformation $w = \sin z$

Introduction

A number of the form $x+iy$ where x and y are real numbers and $i = \sqrt{-1}$ is called a complex number. The set of complex number is denoted By C.

If $z = x+iy$ is a complex number, then x is called the real part of z and we write $Re(z) = x$, y is called the imaginary part of z and we write $Im(z) = y$.

For every complex number $z = x+iy$ there is another complex number called conjugate of z and is defined and denoted by $\bar{z} = x - iy$.

The modulus of z is defined as $r = |z| = \sqrt{x^2+y^2}$ and the amplitude or argument of z as $\theta = \arg z = \tan^{-1}(y/x)$

Let r and θ be polar coordinates of the point (x,y). Then $x = r\cos\theta$ $y = r\sin\theta$ and $z = x+iy = r\cos\theta + ir\sin\theta = r(\cos\theta + i\sin\theta)$ i.e. the polar form of a complex number $z = re^{i\theta}$

Also $\bar{z} = r(\cos\theta - i\sin\theta) = re^{-i\theta}$

Results

$$\boxed{\begin{aligned} |z|^2 &= z\bar{z} \\ Re(z) &= \frac{z+\bar{z}}{2} \\ Im(z) &= \frac{z-\bar{z}}{2i} \end{aligned}}$$

Functions of a Complex Variable

If x and y are real variables, then $z = x+iy$ is called a complex variable. If corresponding to each value of a complex variable z in a given region R, there corresponds one or more values of another complex variable $w = u+iv$, then w is called a function of the complex variable z and is denoted by $w = f(z) = u+iv$.

If to each value of z, there corresponds one and only one value of w, then w is called a single valued function of z otherwise a multi-valued function.

eg: $w = z^2$ $\quad$ $w = \cos z$ $\quad$ $w = e^z$

By separating into real and imaginary parts any function $f(z)$ can be expressed in terms of two real functions as

$$\boxed{w = f(z) = u(x,y) + i v(x,y)}$$

<u>Limit</u>

A function $w = f(z)$ is said to have the limit l as z approaches a point z_0, written

$$\boxed{\lim_{z \to z_0} f(z) = l}$$

OR

~~If~~ A function $w = f(z)$ is defined in a neighborhood of z_0 and if the values of f are close to l for all z close to z_0 if for every positive real ϵ we can find a positive real δ such that for all $z \neq z_0$ in the disk $|z - z_0| < \delta$ we have $|f(z) - l| < \epsilon$.

Continuity

A function $f(z)$ is said to be continuous at $z = z_0$ if $f(z_0)$ is defined and

$$\boxed{\lim_{z \to z_0} f(z) = f(z_0)}$$

$f(z)$ is said to be continuous in a domain if it is continuous at each point of this domain.

Derivative of $f(z)$

The derivative of a complex function f at a point z_0 is written as $f'(z_0)$ and is defined by

$$\boxed{f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}}$$

Provided this limit exists. Then f is said to be differentiable at z_0.

If $\Delta z = z - z_0 \Rightarrow z = z_0 + \Delta z$ then

$$\boxed{f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}}$$

Problems

1. Does the limit $\lim_{z\to 0} \frac{z}{\bar{z}}$ exist. If yes so find the value. ~~of~~ If no explain why?

Limit $\Rightarrow \lim_{z\to z_0} f(z) = l$

Let $z\to 0$ along the path $y = mx$

$z\to 0 \Rightarrow x\to 0$ and $y\to 0$.

$$\lim_{z\to 0} f(z) = \lim_{z\to 0} \frac{z}{\bar{z}}$$

$$= \lim_{z\to 0} \frac{x+iy}{x-iy}$$

$$= \lim_{x\to 0} \frac{x+imx}{x-imx}$$

$$= \lim_{x\to 0} \frac{\cancel{x}(1+im)}{\cancel{x}(1-im)}$$

$$= \frac{1+im}{1-im}$$, which depends on the value of m.

$\therefore \lim_{z\to 0} f(z)$ does not exist

2. Show that $\lim_{z\to 0} \frac{x^2y}{x^4+y^2}$ does not exist.

$$\lim_{z\to z_0} f(z) = l$$

Let $z\to 0$ along $y = mx^2$

$$\lim_{z \to 0} f(z) = \lim_{x \to 0} \frac{x^2 y}{x^4 + y^2}$$

$$= \lim_{x \to 0} \frac{x^2 m x^2}{x^4 + m^2 x^4}$$

$$= \lim_{x \to 0} \frac{\cancel{x^4} \cdot m}{\cancel{x^4}(1+m^2)}$$

$$= \frac{m}{1+m^2}$$, which depends on

the value of m.

$\therefore$ the limit does not exist.

3. Show that the function $f(z) = \begin{cases} \frac{\operatorname{Im}(z)}{|z|}, & z \neq 0 \\ 0 & z = 0 \end{cases}$

is not continuous at $z=0$

$$\lim_{z \to z_0} f(z) = f(z_0)$$

Consider the path $y = mx$

$$\lim_{z \to 0} \frac{\operatorname{Im}(z)}{|z|} = \lim_{x \to 0} \frac{y}{\sqrt{x^2+y^2}}$$

$z = x+iy$
$\operatorname{Im}(z) = y$
$|z| = \sqrt{x^2+y^2}$

$$= \lim_{x \to 0} \frac{mx}{\sqrt{x^2+m^2x^2}}$$

$$= \lim_{x \to 0} \frac{m\cancel{x}}{\cancel{x}\sqrt{1+m^2}}$$

$$= \frac{m}{\sqrt{1+m^2}}$$, which depends on m.

Hence the limit does not exist at $z=0$.

$\therefore$ the function is not continuous at $z=0$.

4. Show that the function $f(z) = \begin{cases} \frac{Re(z^2)}{|z|^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

is not continuous at $z = 0$.

$$\lim_{z \to z_0} f(z) = f(z_0)$$

Consider the path $y = mx$

$$\lim_{z \to 0} f(z) = \lim_{z \to 0} \frac{Re(z^2)}{|z|^2}$$

$$= \lim_{z \to 0} \frac{x^2 - y^2}{x^2 + y^2}$$

$$= \lim_{x \to 0} \frac{x^2 - m^2x^2}{x^2 + m^2x^2}$$

$$= \lim_{x \to 0} \frac{x^2(1 - m^2)}{x^2(1 + m^2)}$$

$$= \frac{1 - m^2}{1 + m^2}, \text{ which depends on } m.$$

$z^2 = (x + iy)^2$

$= x^2 + 2ixy + (iy)^2$

$= (x^2 - y^2) + i2xy$

$Re(z^2) = x^2 - y^2$

$|z|^2 = x^2 + y^2$

Hence limit does not exist at $z = 0$.

$\therefore$ the function is not continuous at $z = 0$.

5. Check whether the function $f(z) = \begin{cases} \frac{Re(z^2)}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

is continuous at $z = 0$.

$$\lim_{z \to z_0} f(z) = f(z_0)$$

Consider the path $y = mx$

$$\lim_{z \to 0} f(z) = \lim_{z \to 0} \frac{x^2 - y^2}{\sqrt{x^2 + y^2}}$$

$$= \lim_{x \to 0} \frac{x^2 - m^2x^2}{\sqrt{x^2 + m^2x^2}}$$

$$= \lim_{x \to 0} \frac{x^{\cancel{2}}(1 - m^2)}{\cancel{x}\sqrt{1 + m^2}} = 0 = f(0)$$

$Re(z^2) = x^2 - y^2$

$|z| = \sqrt{x^2 + y^2}$

$\therefore \lim_{z \to 0} f(z) = f(0) \implies f(z)$ is continuous at $z = 0$

6. Show that $f(z) = \begin{cases} \frac{z\,Re(z)}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

is not differentiable at $z=0$.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

$$f'(0) = \lim_{z \to 0} \frac{f(z) - f(0)}{z} \qquad [\text{Here } z_0 = 0]$$

Let $z \to 0$ along $y = mx$

$$f'(0) = \lim_{z \to 0} \left[\frac{\frac{\cancel{(x+iy)}\, x}{\sqrt{x^2+y^2}} - 0}{\cancel{x+iy}} \right]$$

$z = x+iy$

$Re(z) = x$

$|z| = \sqrt{x^2+y^2}$

$f(0) = 0$

$$= \lim_{z \to 0} \frac{x}{\sqrt{x^2+y^2}}$$

$$= \lim_{x \to 0} \frac{x}{\sqrt{x^2+m^2x^2}}$$

$$= \lim_{x \to 0} \frac{\cancel{x}}{\cancel{x}\sqrt{1+m^2}} = \frac{1}{\sqrt{1+m^2}},$$

which depends on m.

$\therefore$ the limit does not exist.

$\therefore$ $f(z)$ is not differentiable at $z=0$.

7. Show that the function $f(z) = \sqrt{|xy|}$ is not differentiable at origin.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

$$f'(0) = \lim_{z \to 0} \frac{f(z) - f(0)}{z}$$

Let $z \to 0$ along $y = mx$

$$f'(0) = \lim_{z \to 0} \frac{\sqrt{|xy|} - f(0)}{x+iy}$$

$$= \lim_{x \to 0} \frac{\sqrt{|x\, mx|}}{x + imx}$$

$$= \lim_{x \to 0} \frac{x\sqrt{|m|}}{x(1+im)}$$

$$= \frac{\sqrt{|m|}}{1+im}$$, depends on the value of m.

Hence limit does not exist.

$\therefore$ $f(z)$ is not differentiable at origin

8. Prove that the function $f(z) = \frac{x^2 y^5 (x+iy)}{x^4 + y^{10}}$, $z \neq 0$ and $f(0) = 0$ is not differentiable at origin.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

$$f'(0) = \lim_{z \to 0} \frac{f(z) - f(0)}{z}$$

Let $z \to 0$ along the path $y^5 = mx^2$

$$f'(0) = \lim_{z \to 0} \frac{\frac{x^2y^5(x+iy)}{x^4+y^{10}} - 0}{x+iy}$$

$$= \lim_{z \to 0} \frac{x^2y^5}{x^4+y^{10}}$$

$\boxed{y^{10} = (y^5)^2}$

$$= \lim_{x \to 0} \frac{x^2(mx^2)}{x^4 + (mx^2)^2}$$

$$= \lim_{x \to 0} \frac{mx^4}{x^4 + m^2x^4}$$

$$= \lim_{x \to 0} \frac{m\cancel{x^4}}{\cancel{x^4}(1+m^2)}$$

$= \frac{m}{1+m^2}$, which depends on the values of m.

Hence limit does not exist.

$\therefore f(z)$ is <u>not</u> differentiable at $z=0$

Analytic Functions (Holomorphic function, Regular function)

A function $f(z)$ is said to be analytic in a domain D if $f(z)$ is defined and differentiable at all points of D.

The function $f(z)$ is said to be analytic at a point $z = z_0$ in D if $f(z)$ is analytic in a neighbourhood of z_0.

eg: e^z,

The function $f(z) = z^2$ is differentiable only at $z=0$ and it is not differentiable in the neighbourhood of $z_0 = 0$.

$f(z) = z^2$. not analytic function.

Cauchy - Riemann Equations

U.Q. Let $f(z) = u(x,y) + iv(x,y)$ be defined and continuous in some neighbourhood of a point $z = x+iy$ and differentiable at z itself. Then prove that the first order partial derivatives of u and v exist and satisfy the Cauchy Riemann equations, $u_x = v_y$ and $u_y = -v_x$

OR

Show that if $f(z) = u(x,y) + iv(x,y)$ is analytic, then $u(x,y)$ and $v(x,y)$ satisfy Cauchy Riemann equations $u_x = v_y$ and $u_y = -v_x$

Proof

Let $f(z) = u(x,y) + iv(x,y)$ be differentiable at any point z, then

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} \text{ exist.}$$

$$= \lim_{\Delta z \to 0} \left\{ \frac{u(x+\Delta x, y+\Delta y) + iv(x+\Delta x, y+\Delta y) - [u(x,y) + iv(x,y)]}{\Delta x + i\Delta y} \right\}$$

$$= \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \left\{ \left[\frac{u(x+\Delta x, y+\Delta y) - u(x,y)}{\Delta x + i\Delta y}\right] + i\left[\frac{v(x+\Delta x, y+\Delta y) - v(x,y)}{\Delta x + i\Delta y}\right] \right\} \quad \text{--- (1)}$$

If $\Delta z \to 0$ along a line parallel to the real axis $\Delta y = 0$

$\therefore \Delta z = \Delta x$ and hence

$$f'(z) = \lim_{\Delta x \to 0} \left\{ \frac{u(x+\Delta x, y) - u(x,y)}{\Delta x} + i\frac{v(x+\Delta x, y) - v(x,y)}{\Delta x} \right\}$$

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \quad \text{--- (2)}$$

If $\Delta z \to 0$ along a line parallel to the imaginary axis $\Delta x = 0$

$\therefore \Delta z = i\Delta y$ and hence

$$f'(z) = \lim_{\Delta y \to 0}\left[\frac{u(x,y+\Delta y)-u(x,y)}{i\Delta y} + i\,\frac{v(x,y+\Delta y)-v(x,y)}{i\Delta y}\right]$$

$$= \frac{1}{i}\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

$$f'(z) = -i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \quad \text{--- (3)}$$

From (2) and (3)

$$\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

Equating real and imaginary parts

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}}$$

or $u_x = v_y$ and $u_y = -v_x$, Cauchy ~~Riemann~~ Riemann equation.

Note

I. Sufficient condition

The complex function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D of the complex plane, if

(1) u, v are differentiable in D and $u_x = v_y$ and $u_y = -v_x$.

(2) The partial derivatives u_x, u_y, v_x, v_y are all continuous in D.

II The derivative of $f(z)$ is

$$\boxed{f(z) = u_x + i v_x}$$

Problems

1\. Show that $f(z) = e^z = e^{x+iy}$ is analytic. Find $f(z)$.

$f(z) = e^z = e^{x+iy} = e^x \cdot e^{iy}$

$u + iv = e^x [\cos y + i \sin y] = e^x \cos y + i e^x \sin y$

$u = e^x \cos y \qquad v = e^x \sin y$

$u_x = e^x \cos y \qquad v_x = e^x \sin y$

$u_y = e^x(-\sin y) \qquad v_y = e^x \cos y$

CR eqn $u_x = v_y$ and $u_y = -v_x$

$\therefore$ C.R eqn satisfied.

$\therefore$ $f(z)$ is analytic

$f'(z) = u_x + i v_x = e^x \cos y + i e^x \sin y$

$= e^x (\cos y + i \sin y) = e^x \cdot e^{iy}$

$= e^{x+iy} = e^z$

2\. Prove that the function $\sin z$ is analytic and find its derivative.

$f(z) = \sin z = \sin(x+iy)$

$= \sin x \cos(iy) + \cos x \sin(iy)$

$u+iv = \sin x \cosh y + i\cos x \sinh y$

$\sin(A+B) =$ $\sin A \cos B + \cos A \sin B$
$\sin ix = i\sinh x$ $\cos ix = \cosh x$
$\frac{d}{dx}(\sinh x) = \cosh x$ $\frac{d}{dx}(\cosh x) = \sinh x$
$\cos(A+B) =$ $\cos A \cos B - \sin A \sin B$

$u = \sin x \cosh y$ $\quad V = \cos x \sinh y$

$u_x = \cos x \cosh y$ $\quad V_x = -\sin x \sinh y$

$u_y = \sin x \sinh y$ $\quad V_y = \cos x \cosh y$

$u_x = V_y$ and $u_y = -V_x$

C.R. equation Satisfied.

$\therefore f(z)$ is analytic.

$\therefore f'(z) = u_x + iV_x$

$= \cos x \cosh y + i(\sin x \sinh y)$

$= \cos x \cos(iy) - i\sin x \frac{1}{i} \sin(iy)$

$= \cos x \cos(iy) - \sin x \sin(iy)$

$= \cos(x+iy)$

$= \cos z$

3. Prove that the function $\cos z$ is differentiable and find its derivative.

$f(z) = \cos z = \cos(x+iy)$

$= \cos x \cos(iy) - \sin x \sin(iy)$

$u+iV = \cos x \cosh y - i\sin x \sinh y$

$u = \cos x \cosh y$ $\qquad$ $v = -\sin x \sinh y$

$u_x = -\sin x \cdot \cosh y$ $\qquad$ $v_x = -\cos x \sinh y$

$u_y = \cos x \sinh y$ $\qquad$ $v_y = -\sin x \cosh y$

$u_x = v_y$ and $u_y = -v_x$

C-R equation satisfied

$\therefore f(z)$ is differentiable.

$f'(z) = u_x + iv_x = -\sin x \cosh y + i\, -\cos x \sinh y$

$= -\left[\sin x \cos(iy) + i \cos x \frac{1}{i} \sin iy\right]$

$= -\left[\sin x \cos iy + \cos x \sin iy\right]$

$= -\sin(x+iy) = \underline{\underline{-\sin z}}$

4. Prove that the function $\sinh z$ is analytic and find its derivative.

$$\boxed{\begin{aligned}&\sin(A-B) = \\ &\sin A \cos B - \\ &\cos A \sin B\end{aligned}}$$

$f(z) = \sinh z = \frac{1}{i} \sin(iz)$

$= \frac{1}{i}\left[\sin i(x+iy)\right] = \frac{1}{i} \sin(ix - y)$

$$\boxed{\frac{1}{i} = -i}$$

$= \frac{1}{i}\left[\sin(ix) \cos y - \cos(ix) \sin y\right]$

$= \frac{1}{i}\left[i \sinh x \cos y - \cosh x \sin y\right]$

$u + iv = \sinh x \cos y + i \cosh x \sin y$

$u = \sinh x \cos y$ $\qquad$ $v = \cosh x \sin y$

$u_x = \cosh x \cos y$ $\qquad$ $v_x = \sinh x \sin y$

$u_y = \sinh x \, \bar{\sin} y$ $\qquad$ $v_y = \cosh x \cos y$

$u_x = v_y$ and $u_y = -v_x$

C-R. equations Satisfied.

$\therefore$ $f(z)$ is analytic.

$$f'(z) = u_x + i v_x = \cosh x \cos y + i \sinh x \sin y$$

$$= \cos(ix) \cos y + \sin(ix) \sin y$$

$$= \cos(ix - y)$$

$$= \cos i\left(x - \tfrac{1}{i} y\right) = \cos i(x + iy)$$

$$= \cos iz = \underline{\underline{\cosh z}}$$

5. Verify Cauchy-Riemann equation for the function analytic. $f(z) = \cosh z$. Are the function analytic.

$$f(z) = \cosh z = \cancel{\cos i(x+iy)} \cdot \cos iz = \cos i(x+iy)$$

$$= \cos(ix - y)$$

$$= \cos(ix) \cos(y) + \sin(ix) \sin(y)$$

$\boxed{\cos(A-B) = \cos A \cos B + \sin A \sin B}$

$$u + iv = \cosh x \cos y + i \sinh x \sin y$$

$u = \cosh x \cos y$ $\qquad$ $v = \sinh x \sin y$

$u_x = \sinh x \cos y$ $\qquad$ $v_x = \cosh x \sin y$

$u_y = \cosh x \, \bar{\sin} y$ $\qquad$ $v_y = \sinh x \cos y$

$u_x = v_y$ and $u_y = -v_x$. C-R equation Satisfied.

$\therefore$ $f(z)$ analytic.

6. Show that $f(z) = \log z$ is differentiable except at $z=0$ and find its derivative.

$f(z) = \log z$

$u+iv = \log(x+iy)$

$e^{u+iv} = x+iy$

$\Rightarrow x+iy = e^{u} \cdot e^{iv}$

$x+iy = e^{u}[\cos v + i\sin v]$

$x = e^{u}\cos v \qquad y = e^{u}\sin v$

$x^2+y^2 = (e^{u}\cos v)^2 + (e^{u}\sin v)^2 = e^{2u}\cos^2 v + e^{2u}\sin^2 v$

$= e^{2u}[\cos^2 v + \sin^2 v] = e^{2u}$

$\log(x^2+y^2) = 2u$

$u = \frac{1}{2}\log(x^2+y^2)$

$\frac{y}{x} = \frac{e^{u}\sin v}{e^{u}\cos v} = \tan v$

$\boxed{\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}}$

$\Rightarrow v = \tan^{-1}(y/x)$

$\therefore$

$u = \frac{1}{2}\log(x^2+y^2)$

$u_x = \frac{1}{\cancel{2}}\frac{1}{x^2+y^2}\cancel{2}x = \frac{x}{x^2+y^2}$

$u_y = \frac{1}{\cancel{2}}\frac{1}{x^2+y^2}\cancel{2}y$

$= \frac{y}{x^2+y^2}$

$v = \tan^{-1}(y/x)$

$v_x = \frac{1}{1+y^2/x^2}(y \cdot -1/x^2)$

$= \frac{1}{\frac{x^2+y^2}{x^2}} \cdot \frac{-y}{x^2} = \frac{-y}{x^2+y^2}$

$v_y = \frac{1}{1+y^2/x^2}(1/x)$

$= \frac{1}{\frac{x^2+y^2}{x}} \frac{1}{x} = \frac{x}{x^2+y^2}$

$\therefore u_x = v_y$ and $u_y = -v_x$.

$\therefore f(z)$ is differentiable.

$$f'(z) = u_x + i v_x = \frac{x}{x^2+y^2} - \frac{iy}{x^2+y^2} = \frac{x-iy}{x^2+y^2} = \frac{\bar{z}}{|z|^2}$$

$$= \frac{\bar{z}}{z\bar{z}} = \underline{\underline{1/z}}.$$

7. Show that $f(z) = z^2$ is differentiable.

$$f(z) = z^2 = (x+iy)^2 = x^2 - y^2 + i\,2xy$$

$\therefore u = x^2 - y^2 \qquad v = 2xy$

$u_x = 2x \qquad v_x = 2y$

$u_y = -2y \qquad v_y = 2x$

$u_x = v_y$ and $u_y = -v_x$

C.R. eqn satisfied $\Longrightarrow f(z)$ is differentiable

8. Show that the function $f(z) = e^{y+ix}$ not analytic

$$f(z) = e^{y+ix} = e^y e^{ix} = e^y [\cos x + i \sin x]$$

$u + iv = e^y \cos x + i e^y \sin x$

$u = e^y \cos x \qquad v = e^y \sin x$

$u_x = e^y (-\sin x) \qquad v_x = e^y \cos x$

$u_y = e^y \cos x \qquad v_y = e^y \sin x$

$u_x \neq v_y \qquad u_y \neq -v_x$

$\therefore$ C-R equation not satisfied

$\therefore$ $\underline{f(z) \text{ is not analytic}}$

Properties of analytic function

1. Show that an analytic function with a constant real part is a constant.

Let $f(z) = u+iv$ be analytic. [$\therefore$ C-R eqn satisfied]

u = constant $\Rightarrow u_x = u_y = 0$.

$\therefore$ C-R eqn $\left.\begin{array}{l} u_x = v_y = 0 \\ u_y = -v_x = 0 \end{array}\right\} \Rightarrow v_x = v_y = 0$

$\therefore$ v is constant.

$\therefore$ $\underline{\underline{f(z) = u+iv}}$ is constant

2. Show that an analytic function $f(z) = u+iv$ is constant if its imaginary part is constant.

$f(z) = u+iv$ be analytic $\Rightarrow$ C-R equation satisfied.

V = constant $\Rightarrow V_x = V_y = 0$

$\therefore$ C-R eqn $\left.\begin{array}{l} V_x = -u_y = 0 \\ V_y = u_x = 0 \end{array}\right\} \Rightarrow u_x = u_y = 0$

$\Rightarrow u$ is constant

$\therefore$ $f(z) = u+iv$ = constant

3. Show that an analytic function $f(z) = u+iv$ is constant if its modulus is constant.

Let $f(z) = u+iv$ be analytic with $|f(z)|^2 = u^2+v^2 =$ constant.

$\Rightarrow u^2+v^2 = c$

Differentiating partially with respect to x and y

$2u\,u_x + 2v\,v_x = 0$

$u\,u_x + v\,v_x = 0$

$u\,u_x - v\,u_y = 0$ —— 1 $\left[\begin{array}{l} \text{C-R eqn} \\ v_x = -u_y \end{array}\right]$

$2u\,u_y + 2v\,v_y = 0$

$u\,u_y + v\,v_y = 0$

$u\,u_y + v\,u_x = 0$ —— 2 [C R eqn $v_y = u_x$]

1 $\times u$ + 2 $v \Longrightarrow$

$u^2 u_x - \cancel{uv}\,u_y + \cancel{uv}\,u_y + v^2 u_x = 0$

$u^2 u_x + v^2 u_x = 0$

$\Longrightarrow (u^2+v^2)\,u_x = 0$

$\Longrightarrow u_x = 0$

Similarly $u_y = 0 \Longrightarrow u$ - constant

$\Longrightarrow u_x = v_y = 0$ and $u_y = -v_x = 0$

$\Longrightarrow v$ constant

$\therefore$ $f(z) = u + iv$ is constant

4 Show that an analytic function is constant if its argument is constant.

Let $f(z) = u + iv$ be analytic $\Longrightarrow$ C-R equation satisfied.

$\arg f(z) = \text{constant}$

$\Longrightarrow \tan^{-1}(v/u) = \text{constant}.$

$\Rightarrow \frac{v}{u} = k \Rightarrow v = ku$

$v_x = k u_x$ and $v_y = k u_y$

Eliminating k.

$v_x = \frac{v_y}{u_y} u_x \quad \left[k = \frac{v_y}{u_y} \right]$

$v_x u_y = u_x v_y$

$(-u_y) u_y = u_x u_x$

$-u_y^2 - u_x^2 = 0$

$\Rightarrow u_x^2 + u_y^2 = 0 \Rightarrow u_x = 0$ and $u_y = 0$

$\therefore u =$ constant.

$\therefore v = ku$ also constant

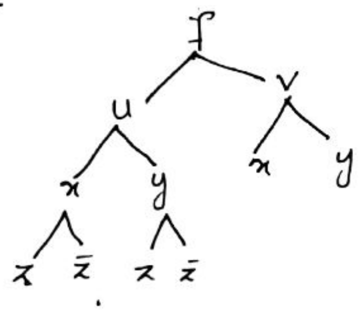
$\therefore f(z) = u + iv$ is constant

5. Prove that an analytic function is independent of $\bar{z}$.

Let $f(z) = u + iv$ be analytic

$\Rightarrow$ C·R eqn $u_x = v_y$ and $u_y = -v_x$.

$x = \frac{z + \bar{z}}{2}$ and $y = \frac{z - \bar{z}}{2i}$



$\frac{\partial f}{\partial \bar{z}} = \frac{\partial u}{\partial \bar{z}} + i \frac{\partial v}{\partial \bar{z}}$

$= \left(\frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \bar{z}} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \bar{z}} \right) +$
$i \left(\frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial \bar{z}} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial \bar{z}} \right)$

$$= \left[u_x \frac{1}{2} + u_y\left(-\frac{1}{2i}\right)\right] + i\left[v_x\left(\frac{1}{2}\right) + v_y\left(-\frac{1}{2i}\right)\right]$$

$$= \frac{1}{2}\left[u_x + iu_y + iv_x - v_y\right]$$

$$= \frac{1}{2}\left[(u_x - v_y) + i(u_y + v_x)\right]$$

$$= \frac{1}{2}\left[(v_y - v_y) + i(-v_x + v_x)\right]$$

$$\frac{\partial f}{\partial \bar{z}} = \underline{\underline{0}} \implies \underline{\underline{f \text{ is independent of } \bar{z}}}$$

6. If $f(z) = u+iv$ is analytic, prove that $u=c_1$ and $v=c_2$ are families of curves cutting orthogonally.

Consider two families of curves $u(x,y) = c_1$ and $v(x,y) = c_2$.

Since u = constant $\implies du = 0$

$$\therefore du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy = 0$$

$$\frac{\partial u}{\partial x}dx = -\frac{\partial u}{\partial y}dy$$

$$\implies \frac{dy}{dx} = \frac{u_x}{-u_y} = m_1$$

Similarly v = constant $\implies dv = 0$

$$\implies m_2 = \frac{-v_x}{v_y}$$

$$\therefore m_1 m_2 = \frac{-u_x}{u_y} \times \frac{-v_x}{v_y} = \frac{-u_x}{u_y} \times \frac{u_y}{u_x} = -1$$

$$m_1 m_2 = -1$$

Hence they are cut orthogonally

Problems

1. Check whether the following functions are analytic or not. Justify your answer.

(i) $f(z) = z + \bar{z}$ (ii) $f(z) = |z|^2$

(i) An analytic function is independent of $\bar{z}$

$f(z) = z + \bar{z}$, which depends on $\bar{z}$

Hence $f(z)$ is not analytic.

(ii) An analytic function is independent of $\bar{z}$

$f(z) = |z|^2 = z\bar{z}$, which depends on $\bar{z}$

Hence $f(z)$ is not analytic

2. Show that $u + iv = \dfrac{x - iy}{x - iy + a}$, $a \neq 0$ is not analytic.

$$u + iv = \frac{x - iy}{x - iy + a} = \frac{\bar{z}}{\bar{z} + a}$$, which depends on $\bar{z}$.

An analytic function is independent of $\bar{z}$

$\therefore$ the given function is not analytic

Harmonic Functions

A real valued function $u(x,y)$ which posses continuous partial derivatives of the first and second order and satisfies Laplace partial differential equations

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad (\text{or } u_{xx} + u_{yy} = 0)$$

is called a harmonic function.

Q: If $f(z) = u+iv$ be an analytic function, prove that the real part and the imaginary part are harmonic functions.

Let $f(z) = u+iv$ be analytic

$\Rightarrow$ C-R equations satisfied.

$u_x = v_y$ and $u_y = -v_x$

$\therefore$ $u_{xx} = v_{xy}$ and $u_{yy} = -v_{xy}$

$\therefore$ $u_{xx} + u_{yy} = v_{xy} - v_{xy} = 0$

Hence $\underline{u \text{ is}}$ harmonic

Similarly $v_{xx} + v_{yy} = -u_{xy} + u_{yx} = 0$

$\Rightarrow$ v is harmonic

Q. Show that $u = \frac{1}{2} \log(x^2+y^2)$ is harmonic

Harmonic $\Rightarrow$ $u_{xx} + u_{yy} = 0$

$u = \frac{1}{2} \log(x^2+y^2)$

$u_x = \frac{1}{2} \frac{1}{x^2+y^2} 2x = \frac{x}{x^2+y^2}$

$u_{xx} = \frac{(x^2+y^2)(1) - x\, 2x}{(x^2+y^2)^2} = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2}$

$u_{xx} = \frac{y^2-x^2}{(x^2+y^2)^2}$

$u_y = \frac{1}{2} \frac{1}{x^2+y^2} 2y = \frac{y}{x^2+y^2}$

$u_{yy} = \frac{(x^2+y^2)(1) - y\, 2y}{(x^2+y^2)^2} = \frac{x^2+y^2-2y^2}{(x^2+y^2)^2}$

$= \frac{x^2-y^2}{(x^2+y^2)^2}$

$u_{xx} + u_{yy} = \frac{y^2-x^2}{(x^2+y^2)^2} + \frac{x^2-y^2}{(x^2+y^2)^2}$

$= \frac{y^2-x^2+x^2-y^2}{(x^2+y^2)^2} = \underline{\underline{0}}$

$\therefore$ u is $\underline{\underline{\text{harmonic}}}$

Q. Determine the constants a and b so that ax^3+bxy is harmonic.

$u = ax^3 + bxy$

u is harmonic $\Longrightarrow u_{xx} + u_{yy} = 0$

$u_x = 3ax^2 + by \qquad u_y = bx$

$u_{xx} = 6ax \qquad u_{yy} = 0$

$u_{xx} + u_{yy} = 0 \Longrightarrow 6ax + 0 = 0$

$\Longrightarrow 6ax = 0$

$\Longrightarrow a = 0$

u is harmonic for any b and $a = 0$

Harmonic Conjugate

If $f(z) = u + iv$ is an analytic function then the real part of $u(x,y)$ and the imaginary part $v(x,y)$ are conjugate harmonic function. In the analytic function 'f', $v(x,y)$ is called the harmonic conjugate of $u(x,y)$.

Problems

1. Verify that $u = x^2 - y^2 - y$ is harmonic in the whole complex plane and find a harmonic conjugate function v of u.

$u = x^2 - y^2 - y$

$u_x = 2x \qquad u_y = -2y - 1$

$u_{xx} = 2 \qquad u_{yy} = -2$

$u_{xx} + u_{yy} = 2 - 2 = 0 \implies$ u is harmonic

Using C-R equation

$v_y = u_x = 2x$ — ① $\qquad v_x = -u_y = 2y + 1$ — ③

Integrating w.r.to y

$v = 2xy + h(x)$ — ②

~~$v_x = -u_y = -(-2y-1) = 2y+1$~~

~~Integrating w.r.to x~~ Diff: ① w.r.to x

$v_x = 2y + h'(x)$ — ④

Com eqn ② and ④

$\cancel{2y} + 1 = \cancel{2y} + h'(x) \longrightarrow h'(x) = 1$

$h(x) = x + c$

$\therefore \; v = 2xy + x + c$

2 Show that $u = \sin x \cosh y$ is harmonic. Find the harmonic conjugate function.

$u = \sin x \cosh y$

$u_x = \cos x \cosh y$ $\qquad$ $u_y = \sin x \sinh y$

$u_{xx} = -\sin x \cosh y$ $\qquad$ $u_{yy} = \sin x \cosh y$

$u_{xx} + u_{yy} = -\sin x \cosh y + \sin x \cosh y = 0$

$\Longrightarrow$ u is harmonic

Using C-R equation

$v_y = u_x = \cos x \cosh y \quad \text{--- (1)}$ $\qquad$ $v_x = -u_y = -\sin x \sinh y \quad \text{--- (2)}$

Integrating (1) w.r.to y

$v = \cos x \sinh y + h(x) \quad \text{--- (3)}$

Diff: (3) w.r.to x

$v_x = -\sin x \sinh y + h'(x) \quad \text{--- (4)}$

Comparing (2) and (4)

$-\sin x \sinh y = -\sin x \sinh y + h'(x)$

$h'(x) = 0$

$h(x) = C$

$\therefore\ v = \underline{\cos x \sinh y} + C$

3. Show that $u = x^3 - 3xy^2$ is harmonic. Find its harmonic conjugate function.

$u = x^3 - 3xy^2$

$u_x = 3x^2 - 3y^2$ $\qquad$ $u_y = -6xy$

$u_{xx} = 6x$ $\qquad$ $u_{yy} = -6x$

$u_{xx} + u_{yy} = 0 \implies 6x - 6x = 0$

$\implies u$ is harmonic

Using Cauchy Riemann equation

$v_y = u_x = 3x^2 - 3y^2$ — (1)

$v_x = -u_y = 6xy$ — (2)

Integrating w. r. to y eqn (1)

$\cancel{V = 6\frac{x^2}{2}y + 1b}$

$V = 3x^2y - \cancel{3}\frac{y^3}{\cancel{3}} + h(x)$ — (3)

Differentiating w. r. to x

$V_x = 3y\,2x + h'(x)$

$V_x = 6xy + h'(x)$ — (4)

Comparing (2) and (4)

$\cancel{6xy} = \cancel{6xy} + h'(x) \implies h'(x) = 0$

Integrate. $h(x) = c$

$\therefore V = 3x^2y - y^3 + c$

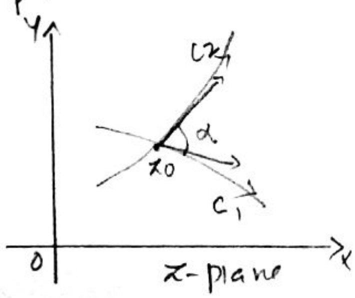
Conformal Mapping

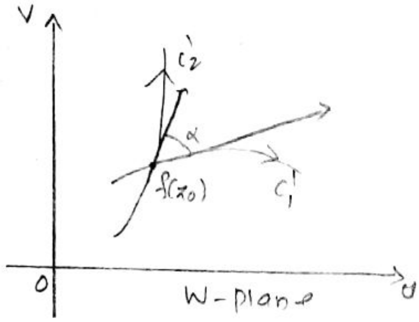
Consider a complex function $w = f(z)$ defined in a domain D of the z-plane, then to each point in D there corresponds a point in the W-plane. This means that there exist a mapping of D onto the range of values of $f(z)$ in the w-plane.

Let $f(z)$ be an analytic function, a point z_0 is called a critical point if the derivative $f'(z)$ is zero at z_0.

The mapping $w = f(z)$ by an analytic function f is conformal, except at critical point.

Suppose two curves C_1, C_2 in the z-plane intersect at the point P and the corresponding curves C_1', C_2' in the w-plane intersect at P' under the transformation $w = f(z)$. If the angle of intersection of the curves at P is the same as the angle of intersection of the curve at P', both in magnitude and sense, then the transformation is said to be conformal at P.





Definition

A transformation which preserves angles both in magnitude and sense between every pair of curves through a point is said to be conformal at the point.

Mapping By elementary functions.

I. The Mapping $w = z^2$

Q: Discuss the transformation $w = z^2$

Let $w = z^2$

$u + iv = (x+iy)^2$

$u + iv = x^2 - y^2 + 2ixy$

$u = x^2 - y^2$, $v = 2xy$

① Consider the vertical line $x = c$, constant

$u = c^2 - y^2$ $\quad$ $v = 2cy$

$y = v/2c$

$\therefore u = c^2 - \frac{v^2}{4c^2}$

$\frac{v^2}{4c^2} = c^2 - u$

$\Rightarrow v^2 = 4c^2(c^2 - u)$, which is a parabola.

② Consider the horizontal line $y = c$, constant

$u = x^2 - c^2$ $\quad$ $v = 2xc$

$x = v/2c$

$$u = \frac{v^2}{4c^2} - c^2$$

$$\frac{v^2}{4c^2} = u + c^2$$

$v^2 = 4c^2(u+c^2)$, which is parabola

(3) The line $x=0$ (imaginary axis)

$u = -y^2 < 0$ and $v=0$.

ie the negative u-axis

(4) The line $y=0$ (real axis)

$u = x^2 > 0$ and $v=0$

ie the positive u-axis

(5) $x^2 - y^2 = c$ (rectangular hyperbola) is mapped on to the line $u=c$ and the rectangular hyperbola $xy=c$ is mapped on to the line $v=2c$.

Problems

1. Find the image of the lines $x=a$, and $y=b$ where 'a' and 'b' are constants, under the transformation $w=z^2$.

$$W = z^2 = (x+iy)^2$$

$$u+iv = x^2 - y^2 + i2xy$$

$$u = x^2 - y^2 \qquad v = 2xy.$$

The line $x = a$

$u = a^2 - y^2$. $v = 2ay$

$u = a^2 - \frac{v^2}{4a^2}$ $y = v/2a$

$\frac{v^2}{4a^2} = a^2 - u$

$\underline{\underline{v^2 = 4a^2(a^2 - u)}}$, represent parabola.

The line $y = b$

$u = x^2 - b^2$ $v = 2xb$

$x = v/2b$

$u = \frac{v^2}{4b^2} - b^2$

$\frac{v^2}{4b^2} = u + b^2$

$\underline{\underline{v^2 = 4b^2(u + b^2)}}$, represent parabola.

2. Under the transformation $w = z^2$, find the image of the triangular region bounded by $x = 1$, $y = 1$ and $x + y = 1$

Let $w = z^2$

$u + iv = (x + iy)^2$

$= x^2 - y^2 + i2xy$

$u = x^2 - y^2$ $v = 2xy$

The line $x = 1$

$u = 1 - y^2$ $v = 2y$

$u = 1 - v^2/4$ $y = v/2$

$\frac{v^2}{4} = 1-u$

$\underline{\underline{v^2 = 4(1-u)}}$, represent parabola.

The line $y=1$

$u = x^2 - 1$ $\qquad v = 2x \Rightarrow x = v/2$

$u = \frac{v^2}{4} - 1 \Longrightarrow \frac{v^2}{4} = u+1$

$v^2 = \underline{4(u+1)}$, represent parabola

The line $x+y=1$

$u = x^2 - y^2 = (x+y)(x-y) = x-y$

$v = 2xy$

$(x+y)^2 = (x-y)^2 + 4xy$

$1 = u^2 + 2v$

$\therefore u^2 = \underline{\underline{1-2v}}$, represent parabola

The image of the region is bounded by parabolas.

3 Find the image of the lines $x=1$, $y=2$ and $x>0$, $y<0$ under the mapping $w=z^2$.

$w = z^2 = (x+iy)^2 = x^2 - y^2 + i\,2xy$

$u = x^2 - y^2$ $\qquad v = 2xy$

The line $x=1$

$u = 1-y^2$ $\qquad v = 2y \Rightarrow y = v/2$

$u = 1 - v^2/4 \Longrightarrow \frac{v^2}{4} = 1-u \Longrightarrow v^2 = 4(1-u)$, represent parabola.

The line $y=2$

$u = x^2 - 4 \qquad v = 4x \qquad x = v/4$

$u = \frac{v^2}{16} - 4 \implies \frac{v^2}{16} = u + 4$

$v^2 = 16(u+4)$ represent parabola

$x > 0, y < 0 \implies 2xy = v < 0$

i.e. image mapped onto the lower half of the plane

4. Find the image of the strip $1/2 \le x \le 1$ under the transformation $w = z^2$.

$w = z^2$

$u + iv = (x+iy)^2 = x^2 - y^2 + 2ixy$

$u = x^2 - y^2 \qquad v = 2xy$

When $x = 1/2$

$u = 1/4 - y^2 \qquad v = y$

$u = \frac{1}{4} - v^2 \implies v^2 = \frac{1}{4} - u$, represent parabola

When $x = 1$

$u = 1 - y^2 \qquad v = 2y \implies y = v/2$

$u = 1 - \frac{v^2}{4} \implies \frac{v^2}{4} = 1 - u$

$v^2 = 4(1-u)$ represent parabola

The image is the region bounded by the parabolas.

5. Find the image of the semi-circle $y = +\sqrt{4-x^2}$ under the transformation $w = z^2$.

$w = z^2$

$u+iv = (x+iy)^2 = x^2-y^2+i2xy$

$u = x^2-y^2 \qquad v = 2xy$

Semi circle $y = +\sqrt{4-x^2}$

$\Rightarrow y^2 = (4-x^2) \Rightarrow x^2+y^2 = 4$

$u^2+v^2 = (x^2-y^2)^2 + 4x^2y^2$

$= x^4+y^4-2x^2y^2+4x^2y^2$

$= x^4+y^4+2x^2y^2$

$= (x^2+y^2)^2$

$= 4^2 = 16$

$\underline{u^2+v^2 = 16}$, represent circle.

II. The Mapping $w = e^z$

Q. Discuss the transformation $w = e^z$.

Let $w = e^z$

$u+iv = e^{x+iy} = e^x \cdot e^{iy} = e^x \cdot [\cos y + i\sin y]$

$u = e^x \cos y \qquad v = e^x \sin y$

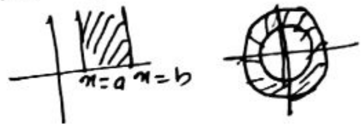
1) The line $x = c$

$u = e^c \cos y \qquad v = e^c \sin y$

$u^2+v^2 = e^{2c}(\cos^2 y + \sin^2 y)$

$u^2+v^2 = e^{2c}$, represent a circle.

2) The strip between $x=a$ and $x=b$ $(a \le x \le b)$ $a \le x \le b$ is transformed to the annular region between the circles $u^2+v^2 = e^{2a}$ and $u^2+v^2 = e^{2b}$.

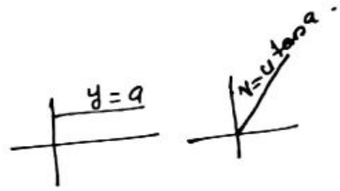


3) When $y=a$, constant

$u = e^x \cos a \qquad v = e^x \sin a$

$$\frac{v}{u} = \frac{e^x \sin a}{e^x \cos a} = \tan a$$

$v = u \tan a$, radial line.



4) The strip $a \le y \le b$ is transformed to the radial $a \le y \le b$ lines $v = u\tan a$ and $v = u \tan b$.

<u>Problems</u>

1. Find the image of $0<x<1$, $\frac{1}{2}<y<1$ under the mapping $w=e^z$.

$W = e^z$

$u+iv = e^{x+iy} = e^x[\cos y + i\sin y]$

$u = e^x \cos y \qquad v = e^x \sin y$

When $0<x<1$

$x=c$ is mapped into the circle $u^2+v^2 = e^{2c}$

$\therefore$

Hence $0 < x < 1$ is mapped into the region between the circles $u^2+v^2 = e^0$ and $u^2+v^2 = e^2$.

When $1/2 < y < 1$

$y = a$ represent a ray $v = u \tan a$.

$\therefore$ $1/2 < y < 1$ represent the region between the radial lines $v = u \tan 1/2$ and $v = u \tan 1$.

$\therefore$ the region is bounded by the concentric circles and rays.

2. Find the image of the region $0 \le x \le 2$, $-\pi \le y \le \pi$ under the mapping $w = e^z$.

~~Let $w = \rho e^{i\phi} = e^z$~~

Let $W = e^z = \rho e^{i\phi}$

$= e^{x+iy} = \rho e^{i\phi}$

$= e^x \cdot e^{iy} = \rho e^{i\phi}$

Comparing $\rho = e^x$ and $\phi = y$.

If $x = 0$, $\rho = e^0 = 1$, which is a circle with radius 1 ~~and~~

$x = 2$ $\rho = e^2$, which is a circle with radius e^2.

Hence $0 \le x \le 2$ is mapped into the region between the circles with radius 1 and e^2.

$y = -\pi$ is mapped on to $\phi = -\pi$ and $y = \pi$ is mapped onto $\phi = \pi$.

III

The transformation $W = 1/z$

Q. Discuss the transformation $W = 1/z$

$$W = \frac{1}{z}$$

$$u + iv = \frac{1}{x+iy} \times \frac{x-iy}{x-iy}$$

$$u + iv = \frac{x-iy}{x^2-(iy)^2} = \frac{x-iy}{x^2+y^2}$$

$$u = \frac{x}{x^2+y^2} \qquad v = \frac{-y}{x^2+y^2}$$

$$\therefore\ x = \frac{u}{u^2+v^2} \qquad y = \frac{-v}{u^2+v^2} \qquad \text{[Rearrange]}$$

1) The line $x = c$

$$\frac{u}{u^2+v^2} = c \implies u = c(u^2+v^2)$$

$\implies c(u^2+v^2) - u = 0$, represent a circle

2) The line $y = c$

$$\frac{-v}{u^2+v^2} = c \implies -v = c(u^2+v^2)$$

$\implies c(u^2+v^2) + v = 0$, represent a circle.

3) The half plane $x > c$

$$\implies \frac{u}{u^2+v^2} > c \iff \frac{u}{c} > u^2+v^2$$

or $u^2+v^2 - \frac{u}{c} < 0$

$\implies (u - 1/2c)^2 + v^2 < (1/2c)^2$, inside the circle.

4) The circle $x^2+y^2=c^2$

$$x^2+y^2=\left(\frac{u}{u^2+v^2}\right)^2+\left(\frac{-v}{u^2+v^2}\right)^2=c^2$$

$$\frac{u^2+v^2}{(u^2+v^2)^2}=c^2$$

$$=\frac{1}{u^2+v^2}=c^2$$

$\Longrightarrow$ $\underline{\underline{u^2+v^2=\frac{1}{c^2}}}$, represent a circle.

Problems

1. Find the image of $|z-1/2|\leq 1/2$ under the transformation $w=1/z$.

$$w=1/z \Longrightarrow z=1/w$$

$$x+iy=\frac{1}{u+iv}\times\frac{u-iv}{u-iv}=\frac{u-iv}{u^2+v^2}$$

$$x=\frac{u}{u^2+v^2} \qquad y=\frac{-v}{u^2+v^2}$$

$$|z-1/2|\leq 1/2 \Longrightarrow (z-1/2)(\overline{z-1/2})\leq 1/4 \qquad \boxed{|z|^2=z\bar{z}}$$

$$\left(z-\frac{1}{2}\right)\left(\bar{z}-1/2\right)=\frac{1}{4}$$

$$\left(\frac{1}{w}-1/2\right)\left(\frac{1}{\bar{w}}-1/2\right)=1/4$$

$$\frac{1}{w\bar{w}}-\frac{1}{2w}-\frac{1}{2\bar{w}}+\frac{1}{4}\leq\frac{1}{4}$$

$$\frac{2-\bar{w}-w}{w\bar{w}}\leq 0 \Longrightarrow 2-\bar{w}-w\leq 0$$

$$2-u+iv-u-iv\leq 0 \Longrightarrow 2-2u\leq 0 \Rightarrow 2\leq 2u$$

$$\underline{u\geq 1}$$

The image is $u \geq 1$

2. Find the image of the region $|z - 1/3| \leq 1/3$ under the transformation $w = 1/z$.

$|z - 1/3| \leq 1/3$

$|z - 1/3|^2 \leq 1/9 \implies (z - 1/3)(\overline{z - 1/3}) \leq \frac{1}{9}$

$(z - 1/3)(\bar{z} - 1/3) \leq \frac{1}{9}$

$\left(\frac{1}{w} - 1/3\right)\left(\frac{1}{\bar{w}} - \frac{1}{3}\right) \leq \frac{1}{9}$

$\frac{1}{w\bar{w}} - \frac{1}{3w} - \frac{1}{3\bar{w}} + \frac{1}{9} \leq 1/9$

$\frac{3 - \bar{w} - w}{3w\bar{w}} \leq 0 \implies 3 - \bar{w} - w \leq 0$

$3 \leq w + \bar{w}$

$3 \leq u + iv + u - iv \implies 3 \leq 2u$

$\implies \frac{3}{2} \leq u$

The image is $u \geq 3/2$

3. Under the transformation $w = 1/z$, find the image of $|z - 2i| = 2$.

$|z - 2i| = 2 \implies |z - 2i|^2 = 4 \implies (z - 2i)(\overline{z - 2i}) = 4$

$(z - 2i)(\bar{z} - 2i) = 4$

$\left(\frac{1}{w} - 2i\right)\left(\frac{1}{\bar{w}} + 2i\right) = 4$

$$\frac{1}{w\bar{w}} + \frac{2i}{w} - \frac{2i}{\bar{w}} + 4 = 4$$

$$\frac{1 + 2i\bar{w} - 2i w}{w\bar{w}} = 0 \longrightarrow 1 + 2i(\bar{w} - w) = 0$$

$\frac{w - \bar{w}}{2i} = v$

$$1 + 2i(-2i v) = 0$$

$1 + 4v = 0$, which is a straight line

Property of the inversion $w = 1/z$

Q. Prove that the mapping $w = 1/z$ maps every straight line or circle onto a circle or straight line.

Every straight line or circle in the z-plane can be written as

$$A(x^2 + y^2) + Bx + Cy + D = 0$$

Where A, B, C, D are real.

If $A = 0$, it gives a straight line and $A \neq 0$ a circle.

In terms of z and $\bar{z}$ the equation becomes

$$A z\bar{z} + B\left(\frac{z + \bar{z}}{2}\right) + C\left(\frac{z - \bar{z}}{2i}\right) + D = 0$$

$$A \frac{1}{w}\frac{1}{\bar{w}} + \frac{B}{2}\left(\frac{1}{w} + \frac{1}{\bar{w}}\right) + \frac{C}{2i}\left(\frac{1}{w} - \frac{1}{\bar{w}}\right) + D = 0$$

$$\frac{A}{w\bar{w}} + \frac{B(\bar{w} + w)}{2w\bar{w}} + \frac{C}{2i}\left(\frac{\bar{w} - w}{w\bar{w}}\right) + D = 0$$

$$\frac{A + \frac{B}{2}(w + \bar{w}) + \frac{C}{2i}(\bar{w} - w) + D w\bar{w}}{w\bar{w}} = 0$$

A

$$A + \frac{B}{2}(w+\bar{w}) + \frac{C}{2i}(\bar{w}-w) + Dw\bar{w} = 0$$

$$A + \frac{B}{2}(2u) + \frac{C}{2i}(-2iv) + D(u^2+v^2) = 0$$

$$A + Bu - Cv + D(u^2+v^2) = 0$$

The equation represent a circle $D \neq 0$ on a straight line if $D=0$ in the w-plane

Fixed points

The points that are mapped onto themselves for the mapping $w=f(z)$ are called fixed points and they are obtained from $w=f(z)=z$.

Problems

I. Find the fixed points of the following mappings.

1. $w = (4+i)z$

Fixed points $\Rightarrow w = f(z) = z$

$\Rightarrow z = (4+i)z$

$\Rightarrow (4+i)z - z = 0 \Longrightarrow (4+i-1)z = 0$

$\Rightarrow \underline{z=0}$

2. $w = (z-i)^2$

Fixed points $w = f(z) = z$

$z = (z-i)^2 \Longrightarrow z = z^2 - 1 - 2iz$

$z^2 - (2i+1)z - 1 = 0$

$$z = \frac{(2i+1) \pm \sqrt{(2i+1)^2 - 4\times 1\times -1}}{2}$$

$$= \frac{(2i+1) \pm \sqrt{\cancel{-4} + 4i + 1 + \cancel{4}}}{2} = \frac{(2i+1) \pm 2\sqrt{i + 1/4}}{2}$$

$$z = \left(i + \tfrac{1}{2}\right) \pm \sqrt{i + 1/4}$$

3. $W = \frac{2iz - 1}{z + 2i}$

$W = f(z) = z \implies z = \frac{2iz-1}{z+2i}$

$z(z+2i) = 2iz - 1$

$z^2 + \cancel{2iz} = \cancel{2iz} - 1$

$z^2 = -1 \implies z = \pm i$

4. $W = \frac{z-1}{z+1}$

fixed points $W = z = \frac{z-1}{z+1}$

$z(z+1) = z - 1$

$z^2 + \cancel{z} = \cancel{z} - 1 \implies z^2 = -1$

$z = \pm i$

II Find the inverse of the following mappings.

(1) $w = \dfrac{3z}{2z-i}$

$w = \dfrac{3z}{2z-i} \Longrightarrow (2z-i)w = 3z$

$2zw - iw = 3z \Longrightarrow 2zw - 3z = iw$

$z(2w-3) = iw$

$z = \dfrac{iw}{2w-3}$

(2) $w = \dfrac{2z+5i}{4z}$

$w = \dfrac{2z+5i}{4z} \Longrightarrow 4zw = 2z+5i$

$4zw - 2z = 5i$

$z(4w-2) = 5i$

$z = \dfrac{5i}{4w-2}$

(3) $w = \dfrac{z-i}{z+i}$

$w(z+i) = z-i$

$wz + wi = z - i$

$wz - z = -wi - i$

$z(w-1) = -iw - i$

$z = \dfrac{-iw - i}{w-1}$

Transformation $w = \sin z$

$w = \sin z$

$u + iv = \sin(x + iy)$

$= \sin x \cos(iy) + \cos x \sin(iy)$

$u + iv = \sin x \cosh y + i \cos x \sinh y$

$u = \sin x \cosh y \qquad v = \cos x \sinh y$

Since $\sin z$ is periodic with period 2π, we restrict to z in the vertical strip $-\pi/2 \le x \le \pi/2$.

The mapping is conformal at all points except at

$f'(z) = 0 \Longrightarrow \cos z = 0 \Longrightarrow z = \pm \pi/2$

① When $x = c$, constant.

$u = \sin c \cosh y \qquad v = \cos c \sinh y$

$\frac{u}{\sin c} = \cosh y \qquad \frac{v}{\cos c} = \sinh y$

$$\cosh^2 y - \sinh^2 y = \frac{u^2}{\sin^2 c} - \frac{v^2}{\cos^2 c}$$

$$1 = \frac{u^2}{\sin^2 c} - \frac{v^2}{\cos^2 c}$$, represent hyperbola

② When $y = c$, constant

$u = \sin x \cosh c \qquad v = \cos x \sinh c$

$\frac{u}{\cosh c} = \sin x \qquad \frac{v}{\sinh c} = \cos x$

$$\sin^2 x + \cos^2 x = \frac{u^2}{\cosh^2 c} + \frac{v^2}{\sinh^2 c}$$

$$1 = \frac{u^2}{\cosh^2 c} + \frac{v^2}{\sinh^2 c}$$, represent ellipse.

(3) When $x = -\pi/2$

$u = \sin(-\pi/2)\cosh y = -\cosh y$ $\qquad$ $v = \cos(-\pi/2)\cosh y$
$\qquad\qquad\qquad\qquad\qquad\qquad = 0$

$u = -\cosh y$ $\qquad$ $\left[\begin{array}{l} \cosh y = \frac{e^{y}+e^{-y}}{2} \geq 1 \\ \cosh y \geq 1 \end{array}\right]$

$u \leq -1$

The image is u-axis such that $u \leq -1$

(4) When $x = \pi/2$

$u = \sin \pi/2 \cosh y = \cosh y \geq 1$ $\qquad$ $v = \cos \pi/2 \cosh y = 0$

The image is the u-axis such that $u \geq 1$.

(5) The semi infinite strip $-\pi/2 \leq x \leq \pi/2$, $y \geq 0$ mapped onto the upper half plane $v \geq 0$.

The right hand strip $0 \leq x \leq \pi/2$, $y \geq 0$ is mapped onto the first quadrant and the left hand strip $-\pi/2 \leq x < 0$, $y \geq 0$ is mapped onto the second quadrant.

Problems

1. Find the image of the region $0 \leq x \leq \pi$, $0 \leq y \leq 1$ under the mapping $w = \sin z$.

$w = \sin z$

$u + iv = \sin(x+iy) = \sin x \cdot \cos(iy) + \cos x \sin(iy)$

$u+iv = \sin x \cdot \cosh y + i \cos x \sinh y$

$u = \sin x \cosh y \qquad v = \cos x \sinh y$

When $0 \leq x \leq \pi$, $\sin x \geq 0$ and hence $u \geq 0$.

If $y = 0$ then $v = 0$.

$y = 1 \quad u = \sin x \cosh 1$ and $v = \cos x \sinh 1$.

$$\frac{u}{\cosh 1} = \sin x \qquad \frac{v}{\sinh 1} = \cos x$$

$$\frac{u^2}{\cosh^2 1} + \frac{v^2}{\sinh^2 1} = \sin^2 x + \cos^2 x$$

$$\Rightarrow \frac{u^2}{\cosh^2 1} + \frac{v^2}{\sinh^2 1} = 1$$

So the image of $0 \leq x \leq \pi$, $0 \leq y \leq 1$ is

$$\frac{u^2}{\cosh^2 1} + \frac{v^2}{\sinh^2 1} \leq 1, \quad u \geq 0.$$

2. Find the image of the region $0 < x < 2\pi$, $1 < y < 5$ under the mapping $w = \sin z$.

$w = \sin z$

$u+iv = \sin(x+iy) = \sin x \cos(iy) + \cos x \sin(iy)$

$u+iv = \sin x \cosh y + i \cos x \sinh y$

$u = \sin x \cosh y \qquad v = \cos x \sinh y$

When $y = c$

$u = \sin x \cosh c \qquad v = \cos x \sinh c$

$$\frac{u}{\cosh c} = \sin x \qquad \frac{v}{\sinh c} = \cos x$$

$$\frac{u^2}{\cosh^2 c} + \frac{v^2}{\sinh^2 c} = \sin^2 x + \cos^2 x = 1$$

$\frac{u^2}{\cosh^2 c} + \frac{v^2}{\sinh^2 c} = 1$, represent ellipse

When $0 \le x \le \pi$, $\sin x \ge 0$ and hence $u \ge 0$.

The image of the segment $0 \le x \le \pi$, $y = c$ is mapped onto the right half of the ellipse.

When $\pi \le x \le 2\pi$, $\sin x \le 0$ and hence $u \le 0$.

The image of $\pi \le x \le 2\pi$, $y=c$ is left half of the ellipse.

When $y = 1$ the ellipse is $\frac{u^2}{\cosh^2 1} + \frac{v^2}{\sinh^2 1} = 1$

When $y = 5$ the ellipse is $\frac{u^2}{\cosh^2 5} + \frac{v^2}{\sinh^2 5} = 1$

$\therefore$ the image of the region $0 < x < 2\pi$, $1 \le y < 5$ is the ellipse annular region bounded by $\frac{u^2}{\cosh^2 1} + \frac{v^2}{\sinh^2 1} = 1$ and $\frac{u^2}{\cosh^2 5} + \frac{v^2}{\sinh^2 5} = 1$

3. Find the image of the region $0 < x < \pi/2$, $0 < y < 2$ under the mapping $w = \sin z$.

$$W = \sin z$$
$$u + iv = \sin(x+iy) = \sin x \cos(iy) + \cos x \sin(iy)$$
$$= \sin x \cosh y + i \cos x \sinh y$$

$u = \sin x \cosh y \qquad v = \cos x \sinh y$

The right hand strip $0 < x < \pi/2$, $y > 0$ is mapped onto the first quadrant.

$y = 2$ mapped onto the ellipse $\frac{u^2}{\cosh^2 2} + \frac{v^2}{\sinh^2 2} = 1$

$\therefore$ the image of the region $0 < x < \pi/2$, $0 < y < 1$ is inside the region of the ellipse in the first quadrant.

QUESTION BANK

1. Show that $f(z) = \sin z$ is analytic for all z. Find $f'(z)$.

2. Show that $f(z) = \begin{cases} \frac{z \operatorname{Re}(z)}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ is not differentiable at $z=0$.

3. Check whether the function $f(z) = \begin{cases} \frac{\operatorname{Re}(z^2)}{|z|^2} & \text{if } z \neq 0 \\ 0 & \text{if } z = 0 \end{cases}$ is continuous at $z=0$.

4. Let $f(z) = u(x,y) + iv(x,y)$ be defined and continuous in some neighbourhood of a point $z = x+iy$ and differentiable at z itself. Then prove that the first order partial derivatives u and v exist and satisfy the Cauchy Rieman equation.

5. Does the limit $\lim_{z \to 0} \frac{z}{\bar{z}}$ exist. If yes find the value. If no explain why?

6. If $f(z) = u+iv$ is analytic, prove that u = constant and v = constant are families of curves cutting orthogonally.

7. Prove that $f(z) = e^{x+iy}$ is analytic. Find $f'(z)$.

8. Check whether the following functions are analytic or not. Justify your answer.

(i) $f(z) = z + \bar{z}$

(ii) $f(z) = |z|^2$

9. Show that $v = 3x^2y - y^3$ is harmonic and find the corresponding analytic function $f(z) = u + iv$

10. Show that $u = y^3 - 3x^2y$ is harmonic and hence find its harmonic conjugate.

11. Define an analytic function and prove that an analytic function of constant modulus is constant.

12. prove that $u = \sin x \cosh y$ is harmonic. Hence find its harmonic conjugate.

13. prove that the function $u(x,y) = x^3 - 3xy^2 - 5y$ is harmonic everywhere. Also find the harmonic conjugate of u.

14. Find the points, if any, in complex plane where the function $f(z) = 2x^2 + y + i(y^2 - x)$ is
   (i) Differentiable (ii) analytic

15. Find an analytic function whose imaginary part is $v(x,y) = \log(x^2+y^2) + x - 2y$.

16. Find the real part of an analytic function whose imaginary part is $v = e^{-x}(x\cos y + y\sin y)$. Also find the corresponding analytic function.

17. Determine the region in the w-plane into which the triangular region bounded by $x=1$ $y=1$ $x+y=1$ is mapped by $w=z^2$.

18. Under the transformation $w=1/z$ find the image of $|z-2i|=2$

19 Find the image of the lines $x=a$ and $y=b$ where a and b are constants under the transforma $w=z^2$

20 Find the image of the semi-circle $y=+\sqrt{4-x^2}$ under the transformation $w=z^2$

21. Find the image of the region $|z-1/3| \leq 1/3$ under the transformation $w=1/z$.

22 Find the image of $0<x<1$, $1/2<y<1$ under the mapping $w=e^z$.

23. Find the image of the rectangular region $-\pi \leq x \leq \pi$, $a \leq y \leq b$ under the mapping $w=\sin z$.

24. Find the image of the lines $x=c$ and $y=k$ where c and k are constants, under the transformation $w=\sin z$.

25. Find the image of $|z-1/2| \leq 1/2$ under the transformation $w=1/z$. Also find the fixed points of the transformation $w=1/z$.